



PFAS exposures and child growth: a longitudinal study from fetal life to early childhood

Marion Ouidir^{a,*}, Marc-André Verner^{b,c}, Valérie Siroux^a, Sarah Lyon-Caen^a, Aminata H. Cissé^d, Jérémie Botton^e, Barbara Heude^d, Séverine Valmary-Degano^f, Line Småstuen Haug^{g,h}, Cathrine Thomsen^{g,h}, Sam Bayat^{i,j}, Rémy Slama^a, Claire Philippat^a

^a University Grenoble Alpes, Inserm U 1209, CNRS UMR 5309, Team of Environmental Epidemiology Applied to Development and Respiratory Health, Institute for Advanced Biosciences, 38000, Grenoble, France

^b Department of Occupational and Environmental Health, School of Public Health, Université de Montréal, C.P. 6128, Succursale Centre-Ville, Montréal, QC, H3C 3J7, Canada

^c Centre de Recherche en Santé Publique, Université de Montréal et CIUSSS du Centre-Sud-de-l'Île-de-Montréal, Montreal, QC, Canada

^d Université Paris Cité and Université Sorbonne Paris Nord, Inserm, INRAE, Center for Research in Epidemiology and Statistics (CRESS), F-75004, Paris, France

^e Université Paris-Saclay, Faculté de Pharmacie, Orsay, 91400, France

^f Biobank BB-0033-00069, Univ. Grenoble Alpes, Inserm U1209, CNRS UMR5309, Institute for Advanced Biosciences, CHU Grenoble-Alpes, F-38000, Grenoble, France

^g Norwegian Institute of Public Health, Department of Food Safety, Oslo, Norway

^h Norwegian Institute of Public Health, Centre for Sustainable Diets, Oslo, Norway

ⁱ Department of Pulmonology and Physiology, Grenoble University Hospital, La Tronche, France

^j Synchrotron Radiation for Biomedicine Laboratory (STROBE), Inserm UA07, Grenoble Alpes University, Grenoble, France

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ABSTRACT

Background: Prenatal exposure to poly- and perfluoroalkyl substances (PFAS) has been associated with lower birth weight or increased adiposity in adolescence. No study has investigated associations with longitudinal growth from conception to early childhood. We explored the association between maternal serum PFAS concentrations during pregnancy and child growth assessed repeatedly from the second trimester of pregnancy to 3 years of age.

Methods: In the SEPAGES cohort, for 450 pregnant women recruited before 19 gestational weeks in Grenoble (France), we measured 26 PFAS from non-fasting maternal serum samples (median gestational age at sampling: 19.4 weeks). Cluster-based analysis identified three PFAS exposure groups (low, moderate, high). Child growth parameters (weight, height, and head parameters) were measured at second and third trimesters (ultrasound examinations), at birth and until 3 years. Using a nonlinear mixed model, we predicted growth parameters and velocities at exactly 3 months and 3 years.

Results: Compared to children belonging to the low PFAS exposure group, those belonging to the high exposure group had higher head circumference during the second trimester (β [95 % CI] = 3.60 [1.49 to 5.72] mm) and at 3 years (39.85 [1.62 to 78.08] mm) as well as higher estimated fetal weight during the second trimester (15.85 [1.48 to 30.21] g) and BMI growth velocity at 3 years (9.66 [1.73 to 17.59] g/m²/month). PFAS concentrations were not associated with growth parameters at third trimester, birth and 3 months.

Conclusions: In this prospective study, maternal serum PFAS concentrations were associated with some child growth parameters, potentially associated with increased risk of obesity in later-life.

1. Introduction

Poly- and perfluoroalkyl substances (PFAS) are synthetic compounds prized for their water and fire repellent properties (Johns and Stead,

2000), widely use in electronic, plastic materials, clothing, and fire-fighting foams. Their stability, earning them the name of “forever chemicals”, combined with extensive use, results in widespread human exposure (De Silva et al., 2021; Sunderland et al., 2019). PFAS have been

* Corresponding author.

E-mail address: marion.ouidir@univ-grenoble-alpes.fr (M. Ouidir).

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detected in maternal blood during pregnancy and in cord blood (Apelberg et al., 2007), indicating that PFAS cross the placental barrier leading to a fetal exposure during the critical prenatal period (Olsen et al., 2009), which is suspected to alter the development and may increase the risk of childhood obesity (Baillie-Hamilton, 2002).

Child obesity has reached epidemic level worldwide (Hu and Staiano, 2022), with higher risk of adulthood obesity, cardiometabolic diseases in later life (Rundle et al., 2020) and reduced life expectancy (Olshansky et al., 2005). Energy imbalance from lack of physical activity and unhealthy diet, does not fully explain this trend (Schwartz et al., 2017).

Prenatal PFAS exposure has been linked to lower birth weight (Gui et al., 2022) and to increased adolescent adiposity (Frigerio et al., 2023) at different age range. Some studies have examined growth trajectories in relation to prenatal PFAS exposure, suggesting potential catch-up growth and variations in weight gain patterns during childhood (Romano et al., 2022; Montazeri et al., 2023; Gao et al., 2022). However, further longitudinal studies are needed to clarify the divergent association sometimes observed between prenatal PFAS exposure and child growth at different ages, that could result from periods of rapid growth, wherein children born smaller due to higher exposure may catch up to or even exceed the weight and adiposity of those born with adequate weight, by an increased growth velocity (Monteiro and Victora, 2005).

Because individuals are exposed to multiple PFAS simultaneously, recent studies have investigated mixture effects using various statistical methods, such as Bayesian Kernel Machine regression (BKMR) (Papadopoulou et al., 2021; Starling et al., 2024), weighted quantile sum approach (WQS) (Gao et al., 2022; Zhang et al., 2022; Li et al., 2021), Bayesian WQS (Knox et al., 2024), quantile g-computation (Zhang et al., 2023a; Beck et al., 2025; Fossa et al., 2025), or elastic-net regressions (Starling et al., 2019; Cai et al., 2023), yielding to divergent conclusions, including associations of PFAS mixtures with decreased childhood adiposity (Papadopoulou et al., 2021; Zhang et al., 2022), increased risk of obesity (Zhang et al., 2023a) or no effect (Li et al., 2021; Cai et al., 2023). Comprehensive studies on PFAS mixture are thus still needed.

Finally, higher glomerular filtration rate (GFR) during pregnancy is associated with higher birth weight (Morken et al., 2014), and lower serum PFAS concentrations due to urinary excretion (Watkins et al., 2013), potentially biasing the association between maternal serum PFAS concentrations and birth weight if not adequately controlled (Verner et al., 2015). To date, few studies have accounted for this potential confounding by kidney function (Sagiv et al., 2017; Taibl et al., 2023; Manzano-Salgado et al., 2017). While previous studies reported a limited influence of GFR on PFAS–birth weight associations, their estimations relied on imperfect equations to compute eGFR (Taibl et al., 2023; Manzano-Salgado et al., 2017), as GFR fluctuate during non-pathological pregnancy (Abduljalil et al., 2012). In some cases blood samples were collected early in pregnancy (Sagiv et al., 2017), when the variation in GFR is less pronounced (Abduljalil et al., 2012). Furthermore, a recent meta-analysis (Steenland et al., 2018) showed that the magnitude of the association between PFAS and birthweight was stronger in studies with blood collected later in pregnancy, supporting a potential role of GFR as a time-dependent confounder. Therefore, we considered it important to explore this source of bias using physiologically based pharmacokinetic (PBPK) modeling tailored to our study context.

We explored the associations between prenatal exposure to a PFAS mixture and child growth parameters from *in-utero* to 3 years of age in the SEPAGES cohort, as well as whether infant's sex serves as an effect modifier of these associations. Because we did not measure kidney function in this cohort, we relied on PBPK modeling (Verner et al., 2015) to evaluate potential confounding by GFR.

2. Methods

2.1. Study population

We relied on the SEPAGES cohort in which 484 pregnant women were recruited in the Grenoble area, France, between July 2014 and July 2017, before 19 gestational weeks (GW) and followed-up (Lyon-Caen et al., 2019). To be included, women had to be more than 18 years of age, with a singleton pregnancy, be enrolled in the French social security system, speak and write French, plan to give birth in one of the four maternity wards of the Grenoble urban areas and to not plan to live outside the Grenoble area within two years.

Both parents gave informed consent to participate. The SEPAGES study was approved by the French ethical committees: Comité de Protection des Personnes Sud-Est V (CPP) and Commission Nationale Informatique et des Libertés (CNIL).

2.2. PFAS exposure assessment

Non-fasting maternal blood samples were collected by trained fieldworkers (median gestational age at sampling: 19.4, 5–95th percentiles: 16.1–28.5 GW) and transported on ice to the biobank of Grenoble Alpes University Hospital to be stored at -80°C . Blood sample processing and PFAS assessment has been already described elsewhere (Philippat et al., 2023). Briefly, aliquots of serum were sent on dry ice to the Norwegian Institute of Public Health for assessment of 26 PFAS (detailed in Excel Table S1) using on-line solid phase extraction with high performance liquid chromatography and mass spectrometry (HPLC-MS-MS) (Haug et al., 2009; Poothong et al., 2017). The relative standard deviations using controls samples included in the sample batches were below 13 % (Philippat et al., 2023). Among the 26 PFAS analyzed, 13 were quantified in less than 5 % of the samples, and were thus excluded (Excel Table S2). PFAS quantified in less than 70 % of the samples (i.e. PFDoDA, PFHpA, PFTrDA, PFBS, PFHxPA, 6:2 diPAP), were categorized as below vs. above the limit of quantification (LOQ). For the remaining PFAS (i.e. PFOA, PFNA, PFDA, PFHxS, PFOS, PFUnDA, PFHpS), concentrations below the LOQ were singly imputed using the compound's probability distribution (Helsel, 2011); then, standardized (for transport time and batch) (Mortamais et al., 2012) and ln-transformed prior to analyses. In a previously published paper, patterns of PFAS exposure concentrations were identified (Philippat et al., 2023) using a model-based clustering approach, specifically the VarSelLCM R package (Marbac and Sedki, 2017). Briefly, based on statistical criteria applied to the ln-transformed PFAS concentrations (i.e. PFOA, PFNA, PFDA, PFHxS, PFOS, PFUnDA, PFHpS), three clusters of volunteers were identified with a high overall mean posterior probability (0.96), indicating strong model performance (Philippat et al., 2023). These clusters identified groups of volunteers with low [$n = 163$], moderate [$n = 236$], and high [$n = 51$] PFAS concentrations (Supplementary Fig. S1).

2.3. Outcomes

During pregnancy, fetal abdominal circumference, biparietal diameter, femoral length, and head circumference were measured (in mm) by ultrasound examinations as a part of the French routine pregnancy follow-up and those corresponding to the second and third ultrasound examinations (median 22 and 32 GW, respectively) were extracted. Estimated fetal weight was calculated (in g) using the Hadlock formula (Hadlock et al., 1985).

At birth, newborns' weight, length, and head circumference were measured by the maternity staff in the delivery room within 2 h of birth and were extracted.

Child weight, height and head circumference were measured during SEPAGES study visits at 2 months, 1 year and 3 years. Additionally, all measurements of these anthropometric parameters performed by health

care practitioners between 0 and 3 years were copied from the child's health booklet. Then, postnatal child growth was individually modelled using the Jenss-Bayley nonlinear mixed model (Botton et al., 2014; Comets et al., 2017; Philippat et al., 2014), for all children having at least two measurements; with the exception of weight measurements in the first 3 days of life, to avoid the physiological neonatal weight loss. The Jenss-Bayley nonlinear mixed model was performed separately for males and females. This method address the potential issue of missing data and loss to follow-up, and allow to calculate postnatal growth velocity by using the first derivative of the individual equation (Botton et al., 2014), for weight (in g/month), height (in cm/months) and body mass index (BMI, in g/m²/month). We predicted the weight, height, head circumference, BMI and their respective growth velocities at 3 months and 3 years because they are predictive of the risk of obesity in later life (Botton et al., 2008; Young et al., 2012), and 3 years was the oldest age with available growth measurements when this work was initiated (Ouidir et al., 2024).

In addition, during the 3-year SEPAGES study visit, trained field-workers measured weight (in kg), subscapular and triceps skinfolds (in mm) with a skinfold caliper, abdominal circumference (in cm) and performed bioelectrical impedance analysis (BIA) with an empty bladder using the BodyStat 1500 MDD (Bodystat, Isle of Man, UK). Using the BIA, weight and skinfold thickness, we calculated the body fat free mass using the equation developed in children (Goran et al., 1996), to estimate percentage of body fat (%) and fat mass index (kg/m²) (Colombini et al., 2024).

2.4. Statistical analysis

To limit the number of comparisons in our main analysis we investigated the association of clusters-based exposure (not individual PFAS), with each growth parameters at each time-points (second trimester of pregnancy [T2], third trimester of pregnancy [T3], birth, 3-months, 3-years) using multiple linear regression. To investigate whether these associations may be sex specific, we further performed sex-stratified analysis, only when p-value of interaction between the cluster variable and sex was <0.10.

All analyses were adjusted for potential confounders (i.e. variable predictors of both PFAS exposure and growth outcomes) as well as predictors of our outcome only. These variables were identified a priori using a directed acyclic graph constructed from the literature: maternal characteristics (age [<35 vs. ≥ 35 years], maternal education [Up to 3 years vs. 4 or more years after high-school], pre-pregnancy body weight [continuous], height [continuous], weight gain at blood sampling [continuous], parity [nulliparous vs. parous], smoking status during pregnancy [yes vs. no]), and child characteristics (sex [males vs. females], breastfeeding duration for postnatal outcomes [binary at 3 months (yes vs. no) or continuous (in weeks) at 3 years], postnatal passive smoking prior to the visit [yes vs. no, computed from the 2-months, 1-year and 3-years visits], and age at measurement [gestational age for fetal and birth outcomes or age at clinical visit, continuous]). The coding of adjustment factors was based on their relationship with the outcomes (Slama and Werwatz, 2005).

Missing values of covariates (maternal height [n = 4], maternal weight gain until blood withdrawn [n = 8], breastfeeding duration [n = 31], breastfeeding at 3 months [n = 31], maternal educational level [n = 2], maternal active smoking status during pregnancy [n = 44], child passive smoking [n = 38], child sex [n = 2] ad for sensitivity analysis paternal weight [n = 62] and height [n = 65]) were imputed using multiple imputation based on fully conditional specification method (Buuren and Groothuis-Oudshoorn, 2011), with 20 iterations.

In sensitivity analyses, to test the robustness of our associations, we further 1) adjusted for paternal anthropometric measures (weight and height), 2) adjusted for maternal dietary patterns (fish, crustaceans, eggs and liver consumption) during pregnancy that may be predictors of PFAS concentrations and child growth, 3) winsorizing each growth

parameters at the 1st and 99th percentiles.

To provide effect estimate for future meta-analysis, we conducted multiple linear regression for individual PFAS and sex-specific regression for growth parameters that showed sex-modified effect with cluster-based exposures. These results, used as a support for interpretation and comparison with previous studies, were not corrected for multiple testing.

Since higher GFR during pregnancy can confound the association between maternal serum PFAS and child birth weight, we estimated the related bias, by performing 250,000 Monte Carlo simulations using a previously defined PBPK model (Verner et al., 2015) to generate a simulated population of pregnant women with varying GFR and PFAS concentration. We subsequently performed linear regressions to assess the relationship between simulated PFAS concentrations and birth weight. To estimate bias, we then compared the effect estimates to those obtained in this study, as detailed by Verner et al. (Verner et al., 2015). We only performed this analysis for PFOS and PFOA, since PBPK models were only available for this two PFAS.

Statistical analyses were conducted with R-4.2.2, with R-codes available in the following gitlab repository: <https://gricad-gitlab.univ-grenoble-alpes.fr/iab-env-epi>. Data can be accessible upon request to the SEPAGES principal investigators (R. Slama and C. Philippat) following the necessary regulatory procedures.

3. Results

Out of the 484 study participants, 450 women had blood collected for PFAS measurement (Fig. 1). The final sample size ranged from 325 (3-year fat mass) to 448 (birth weight) depending on the availability of offspring's growth outcomes.

At inclusion, women were in median 32.2 (25–75th percentiles: 30.0 to 35.2) years old, mostly nulliparous (n = 247, 54.9 %) and did not smoke (n = 387, 86 %) (Table 1). Mother-child pairs included in the study only differed from the non-included ones by maternal height (165 [161 to 169] cm vs. 161 [158 to 169] cm, respectively, Excel Table S3).

Within each measurement time-point, correlation was moderate to high between fetal growth measured during pregnancy or at birth (Spearman's correlation estimate rho ranged from 0.31 to 0.92), low to high at 3 months (rho: −0.12 to 0.79) and 3 years (rho: −0.22 to 0.98) (Supplementary Fig. S2). Between time-point, fetal growth measured during pregnancy or at birth were overall negatively correlated with BMI velocity at 3 months (rho: −0.26 to 0.02) and not correlated with fat mass parameters at 3 years (rho: −0.08 to 0.05). Growth velocities of weight, height and BMI estimated at 3 months were negatively correlated with BMI velocity at 3 years (rho: −0.26 to −0.16).

Regrading maternal serum PFAS concentrations, among the PFAS quantified in less than 70 % of the samples (i.e. PFDoDA, PFHpA, PFTrDA, PFBS, PFHxPA, 6:2 diPAP), the quantification rate was from 6 % of the sample (6:2 diPAP) to 52 % (PFHxPA; Excel Table S2). For other PFAS, the frequencies of quantifications were 83 % for PFHpS, 98 % for PFUnDA and 100 % (PFOA, PFNA, PFDA, PFHxS, PFOS). Spearman pairwise correlations between PFAS were all above 0.23 and several PFAS were highly correlated, with the lowest (rho = 0.23) between PFHxS and PFUnDA (Fig. S3). The highest correlation was observed between PFNA and PFDA (rho = 0.78). PFOS was highly correlated with all PFAS (rho ranged between 0.55 and 0.72). Regarding the cluster analysis, among the 163 children in the low PFAS concentrations, only 5 %–21 % of the mothers (depending of the PFAS) had a concentration above the whole population median concentration, whereas 62 %–70 % in the moderate exposure group (n = 236) and 76 %–96 % in the high exposure group (n = 51, Supplementary Fig. S1).

3.1. Cluster-based PFAS analysis

At second trimester children belonging to the high PFAS exposure group had greater head measurement (biparietal diameter, β [95 % CI]

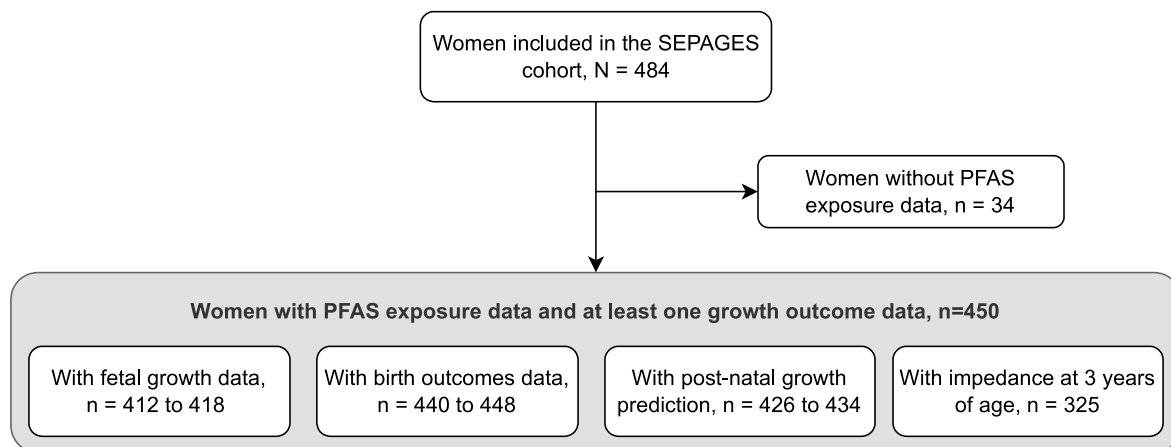


Fig. 1. Flow chart. Abbreviation: PFAS: poly- and perfluoroalkyl substances.

= 1.08 [0.27 to 1.89] mm and head circumference, $\beta = 3.60$ [1.49 to 5.72] mm) and estimated fetal weight ($\beta = 15.85$ [1.48 to 30.21] g), compared to children belonging to the low exposure group (Fig. 2 and Excel Table S4).

PFAS concentrations were not associated with any growth parameters at either third trimester, birth and 3 months.

At 3 years, compared to children belonging to the low exposure group, those in the high exposure group had higher head circumference ($\beta = 39.85$ [1.62 to 78.08] mm and higher BMI growth velocity ($\beta = 9.66$ [1.73 to 17.59] g/m²/month) while children in the moderate exposure group had lower fat mass percent and index ($\beta = -0.92$ [-1.73 to -0.12] % and $\beta = -0.17$ [-0.34 to -0.01] kg/m², respectively).

3.2. Sex-stratified analysis

Cluster-based PFAS exposure suggested effect modifications by child sex for five growth parameters (p-value of interaction <0.10, Table 2). After stratification, a positive association at third trimester with abdominal circumference and estimated fetal weight (high exposure group) emerged in males.

At 3 years, males belonging to the moderate exposure group had a smaller weight ($\beta = -0.62$ [-1.02 to -0.23] kg) and weight velocity ($\beta = -15.32$ [-24.42 to -6.22] g/month), while females belonging to the high exposure group had a higher height ($\beta = 1.67$ [0.10 to 3.23] cm) and weight velocity ($\beta = 21.96$ [2.94 to 40.99] g/month), compared to children belonging to the low exposure group.

3.3. Sensitivity analyses

Regarding sensitivity analysis, further adjustment for paternal anthropometry or dietary patterns yielded similar results (Excel Tables S5 and S6). Analysis using winsorizing maintained most associations: only associations observed in the high concentration group for estimated fetal weight at second trimester was no longer significant ($\beta = 10.19$ [-3.82 to 24.19] g compared to 15.85 [1.48 to 30.21] g in main analysis, Excel Table S7).

3.4. Uni-pollutant analysis

Regarding individual PFAS, most associations with child growth were positive, except 6:2 diPAP that showed negative associations with *in-utero* parameters and PFNA and PFBS that showed negative association with child growth measures at 3 years (Excel Table S8 and detailed results in the supplement). Effect modification by child sex was identified for weight, height and weight velocity, all at 3 years of age (Excel Table S9). Similarly to cluster-based analysis, increased PFAS concentrations tended to lead to smaller males and bigger females at 3 years of

age.

3.5. Estimation of the part of the association with birthweight attributable to GFR

While not statistically significant, in uni-pollutant analysis maternal serum PFOA and PFOS concentrations measured at median 19.4 gestational weeks were associated with 24.90 [-95.76 to 45.96] and 21.10 [-85.74 to 43.55] g decreases in birth weight, respectively. Analysis of the effect attributable to GFR using simulated birth weight and simulated PFAS concentration also showed negative associations (Fig. 3 and Excel Table S10). The increasing negative association observed in the simulated study suggests that the proportion of the PFAS-birth weight association attributable to confounding by GFR increases as gestational age advances (Fig. 3), meaning that studies with blood sample collection later in pregnancy may be more affected by this bias. For example, each ln-increase in simulated maternal serum PFOA and PFOS concentration at 18 gestational weeks was associated with 4.74 [-7.49 to -1.99] and 3.11 [-5.69 to -0.52] g decreases in birth weight, respectively. This suggests that a small proportion (15–20 %) of the effect estimates for birth weight in our study may be attributable to confounding by GFR.

4. Discussion

In this prospective cohort we found associations between mixture of maternal serum PFAS concentrations and children's growth measurements at the second trimester and at 3 years. At both time points children born to women in the high exposure group had higher head measurements, at second trimester fetus had higher fetal weight, while at 3 years children had higher BMI velocity. We did not observe association with growth parameters measured at birth.

Among studies exploring mixture effect, two did not report association between the PFAS mixture and fetal growth ultrasound measurements (Knox et al., 2024; Ouidir et al., 2020), while we reported higher head circumference and higher estimated fetal weight at second trimester. Two other studies included measurement at birth found increased birth weight with prenatal PFAS mixture exposure (Pearce et al., 2021; Law et al., 2023), while we did not observe any association at birth. Regarding post-natal growth, the positive association we observed between PFAS mixture and BMI velocity at 3 years, in line with studies investigating BMI trajectories, with increased odds of belonging to a BMI trajectory of accelerated BMI gain (Montazeri et al., 2023). The velocity increase is a risk of being overweight (Monteiro and Victora, 2005), is in line with results of one study that reported increased risk of overweight in American adolescents with higher maternal plasma PFAS mixture (Zhang et al., 2023a). Regarding marker of adiposity, while some studies found no association (Starling et al., 2024; Li et al., 2021;

Table 1
Characteristics of included mother-child pairs from SEPAGES cohort population (N = 450).

Characteristics	Median [percentile 25, 75] or n (%)
Maternal age at inclusion (years)	32.2 [30.0 to 35.2]
Maternal height (cm) (4 missing)	165.0 [161.0 to 169.3]
Maternal weight before pregnancy (kg) (0 missing)	59.0 [54.0 to 66.1]
Maternal pregnancy weight gain until blood withdrawn (kg) (8 missing)	4.8 [2.8 to 7.1]
Father height (cm) (65 missing)	178.5 [174.2 to 182.3]
Father weight (kg) (62 missing)	76.0 [69.4 to 84.2]
Gestational age at 2nd trimester-ultrasound (weeks) (32 missing)	21.7 [21.3 to 22.1]
Gestational age at 3rd trimester-ultrasound (weeks) (35 missing)	32.4 [32.1 to 32.9]
Gestational age at delivery (weeks) (1 missing)	40.0 [39.0 to 40.7]
Child breastfeeding duration (weeks) (31 missing)	26.0 [13.1 to 43.3]
Parity	
Nulliparous	203 (45.1 %)
Parous	247 (54.9 %)
Maternal education (years after high school)	
Up to 3 years	76 (16.9 %)
4 years or more	372 (82.7 %)
Missing	2 (0.4 %)
Child's sex	
Male	242 (53.8 %)
Female	206 (45.8 %)
Missing	2 (0.4 %)
Child still breastfed at 3 months of life	
Yes	332 (73.8 %)
No	87 (19.3 %)
Missing	31 (6.9 %)
Maternal active smoking at any trimester of pregnancy (above 1 cig./day)	
Yes	19 (4.2 %)
No	387 (86.0 %)
Missing	44 (9.8 %)
Child exposure to tobacco smoke	
Yes	91 (20.2 %)
No	321 (71.3 %)
Missing	38 (8.4 %)
Ultrasound measurements at second trimester (T2)	
Biparietal diameter at T2 (mm) (32 missing)	52.4 [50.5 to 54.3]
Head circumference at T2 (mm) (34 missing)	190.0 [185.0 to 196.6]
Abdominal circumference at T2 (mm) (34 missing)	174.0 [166.8 to 180.5]
Femur length at T2 (mm) (32 missing)	37.2 [35.5 to 38.8]
Estimated fetal weight at T2 (g) (35 missing)	465.1 [424.8 to 508.6]
Ultrasound measurements at third trimester (T3)	
Biparietal diameter at T3 (mm) (35 missing)	82.6 [80.0 to 84.6]
Head circumference at T3 (mm) (38 missing)	294.0 [287.0 to 301.0]
Abdominal circumference at T3 (mm) (36 missing)	285.0 [274.6 to 296.0]
Femur length at T3 (mm) (35 missing)	62.3 [60.8 to 64.0]
Estimated fetal weight at T3 (g) (38 missing)	1955.5 [1825.5 to 2134.9]
Birth measurements	
Birth weight (g) (2 missing)	3280.0 [3037.5 to 3562.5]
Child height at birth (cm) (6 missing)	50.0 [49.0 to 51.5]
Head circumference at birth (cm) (10 missing)	34.5 [34.0 to 35.2]
Growth measurements at 3 months (M3)	
Weight at 3 months (kg) (24 missing)	5.9 [5.5 to 6.3]
Height at 3 months (cm) (16 missing)	60.4 [59.1 to 61.7]
Head circumference at 3 months (cm) (17 missing)	40.5 [39.8 to 41.2]
BMI at 3 months (kg/m ²) (24 missing)	16.0 [15.4 to 16.9]
Weight velocity at 3 months (g/month) (24 missing)	651.5 [589.0 to 753.4]
Height velocity at 3 months (cm/month) (16 missing)	2.7 [2.5 to 2.8]
BMI velocity at 3 months (g/m ² /month) (24 missing)	381.5 [235.2 to 542.8]
Growth measurements at 3 years (Y3)	
Weight at 3 years (kg) (24 missing)	14.0 [13.2 to 15.2]
Height at 3 years (cm) (16 missing)	95.0 [93.0 to 96.8]
Head circumference at 3 years (cm) (17 missing)	49.8 [49.0 to 50.7]
BMI at 3 years (kg/m ²) (24 missing)	15.6 [14.8 to 16.4]
Weight velocity at 3 years (g/month) (24 missing)	187.6 [169.8 to 209.3]

Table 1 (continued)

Characteristics	Median [percentile 25, 75] or n (%)
Height velocity at 3 years (cm/month) (16 missing)	0.7 [0.7 to 0.8]
BMI velocity at 3 years (g/m²/month) (24 missing)	−33.9 [−48.8 to −18.7]
Abdominal circumference at 3 years (cm) (94 missing)	49.0 [47.5 to 51.1]
Fat mass percent at 3 years (%) (125 missing)	13.4 [11.1 to 16.3]
Fat mass index at 3 years (kg/m²) (125 missing)	2.1 [1.7 to 2.7]

Abbreviations: BMI: body mass index.

Cai et al., 2023; Liu et al., 2023), a few reported negative association with waist circumference at 8 years (Papadopoulou et al., 2021), and fat mass at 7 years (Zhang et al., 2022), in line with our result of decreased fat mass in the moderate exposed group compared to lower exposed group. Only one examined postnatal head growth parameters and reported a decrease risk of being in a high-stable or a low-rising trajectory compared to a moderate-stable trajectory (reference) with maternal exposure to a mixture of the same 7 PFAS as in our study (Gao et al., 2022), which is not consistent with our results.

Mixture models used in the study mentioned above included BKMR (Papadopoulou et al., 2021; Starling et al., 2024), WQS (Gao et al., 2022; Zhang et al., 2022; Li et al., 2021), Bayesian WQS (Knox et al., 2024), quantile g-computation (Zhang et al., 2023a; Beck et al., 2025; Fossa et al., 2025), and elastic-net regressions (Starling et al., 2019; Cai et al., 2023), which are not directly comparable with our clustering approach. The mixture models typically estimate regression coefficients (β) representing the average effect of an increase by a given quantile of all chemicals within the mixture. In contrast, our cluster-based approach identifies subgroups of individuals with similar exposure profiles, allowing for comparison between groups characterized by different exposure levels. This method is particularly useful for assessing exposure patterns when individual chemicals are highly correlated, as it is the case for PFAS and when exposure distributions differ substantially across subpopulations.

After sex-stratification, some association emerged, despite the small sample size. The PFAS mixture showed overall negative association among males and positive among females, in line with two recent studies conducted among 7-years old children, one in China (Zhang et al., 2022) and the other in Denmark (Beck et al., 2025). Moreover, one study on growth trajectory reported greater BMI growth observed among females (Romano et al., 2022). Conversely, one study reported negative association between maternal serum PFAS concentrations and weight in the first year of life among British girls (Marks et al., 2021).

Sex-specific associations with PFAS may arise from their effects on the endocrine system, which is by nature sex-specific. PFAS can bind to thyroid hormone receptors (Coperchini et al., 2021), altering levels crucial for fetal brain development and metabolism, leading to growth retardation and developmental delays. They also affect lipid metabolism by activating peroxisome proliferator-activated receptors (PPARs) (Szilagyi et al., 2020), especially PPAR- α , resulting in increased adiposity and weight gain. PFAS induce reactive oxygen species, causing oxidative stress and impairing cell growth (Gundacker et al., 2022). Additionally, PFAS can disrupt placental function (Gan et al., 2024; Chowdhury et al., 2024), impairing nutrient and oxygen transfer to the fetus, which can lead to long-term growth issues.

Considering individual PFAS, the four studies on fetal growth based on ultrasound measurements showed mixed results including no association (Sevelsted et al., 2022), mostly negative associations (Knox et al., 2024; Yang et al., 2025), or, in line with our results, mostly positive associations (Ouidir et al., 2020). In our study, only one PFAS (6:2 diPAP) was negatively associated with fetal growth. However, this result required cautious interpretation since only 6 % of the women had levels higher to the LOQ, and to our knowledge this is the first study investigating prenatal 6:2 diPAP and fetal growth. At birth, while we found few

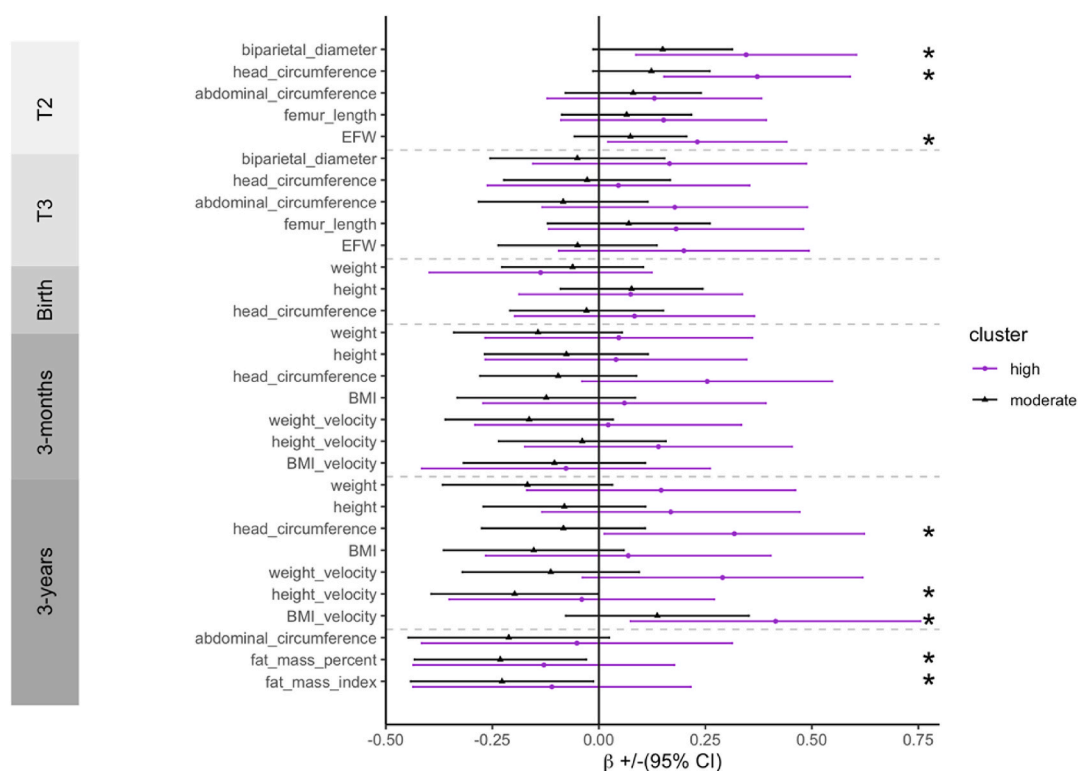


Fig. 2. Adjusted associations between prenatal PFAS exposure patterns (“clusters”) and scaled child growth (N = 450 from the SEPAGES cohort). Clusters representing low (reference), moderate and high PFAS concentration. Abbreviations: BMI: body mass index, EFW: estimated fetal weight, PFAS: poly- and perfluoroalkyl substances.

Table 2

Sex-stratified associations between prenatal PFAS exposure patterns (“clusters”) and child growth (N = 450 from the SEPAGES cohort). Results only displayed for association showing an effect modification by child sex (p-value of interaction between PFAS exposure clusters and child sex <0.10). Clusters representing low (reference), moderate and high PFAS concentration.

Outcomes	Cluster group ^a	Male					Female				
		N	b	[95 %	CI]	p	N	b	[95 %	CI]	p
Abdominal circumference at T3 (mm)	Low (ref.)	76					71				
	Moderate	119	-3.15	-7.43	1.13	0.15	102	1.12	-3.22	5.47	0.61
	High	30	5.44	-0.72	11.60	0.08	16	-1.31	-8.93	6.32	0.74
Estimated fetal weight at T3 (g)	Low (ref.)	76					71				
	Moderate	118	-46.01	-109.45	17.43	0.15	102	29.36	-34.29	93.00	0.36
	High	30	73.15	-18.03	164.34	0.12	15	11.27	-103.57	126.12	0.85
Weight at 3 years (kg)	Low (ref.)	85					70				
	Moderate	118	-0.62	-1.02	-0.23	<0.01	106	0.18	-0.29	0.65	0.45
	High	29	0.02	-0.57	0.61	0.94	18	0.57	-0.23	1.37	0.16
Height at 3 years (cm)	Low (ref.)	86					71				
	Moderate	121	-0.84	-1.67	-0.02	0.05	109	0.53	-0.37	1.44	0.25
	High	29	-0.13	-1.37	1.10	0.83	18	1.67	0.10	3.23	0.04
Weight velocity at 3 years (g/month)	Low (ref.)	85					70				
	Moderate	118	-15.32	-24.42	-6.22	<0.01	106	8.80	-2.31	19.90	0.12
	High	29	3.44	-10.20	17.09	0.62	18	21.96	2.94	40.99	0.02

Abbreviations: BMI: body mass index, PFAS: poly- and perfluoroalkyl substances, T3: third trimester, ref = reference group.

Adjusted for: maternal age, maternal education, maternal pre-pregnancy weight, maternal height, maternal weight gain, parity, maternal smoking status during pregnancy, breastfeeding status for postnatal outcomes, child passive smoking, and age at measurement (gestational age for fetal and birth outcomes, age at clinical visit for measurement at the clinical visit).

negative associations, between prenatal PFNA and birth weight and between prenatal PFDoDA and head circumference, both associations were in line with a recent meta-analysis (Gui et al., 2022). However, the meta-analysis also reported negative associations between birth weight and PFOA, PFOS, PFUnDA and PFDA (Gui et al., 2022), all with a negative trend but not significant in our study. Regarding childhood growth, a recent meta-analysis investigating the effect of 8 PFAS on children adiposity parameters reported either positive associations between prenatal PFNA and PFOA exposures and BMI or negative

associations between prenatal PFOS and PFHxS exposures and, respectively, BMI before 3 years of age or risk of overweight (Frigerio et al., 2023). Those results were not in line with our study, that rather suggested a negative association between PFNA and BMI at 3 years of age and no associations for PFOA, PFOS and PFHxS. However, authors of the meta-analysis, highlighted the high results' heterogeneity across studies (Frigerio et al., 2023).

This study has several strengths and limitations. We could not account for maternal physical activity, folate intake, and maternal GFR

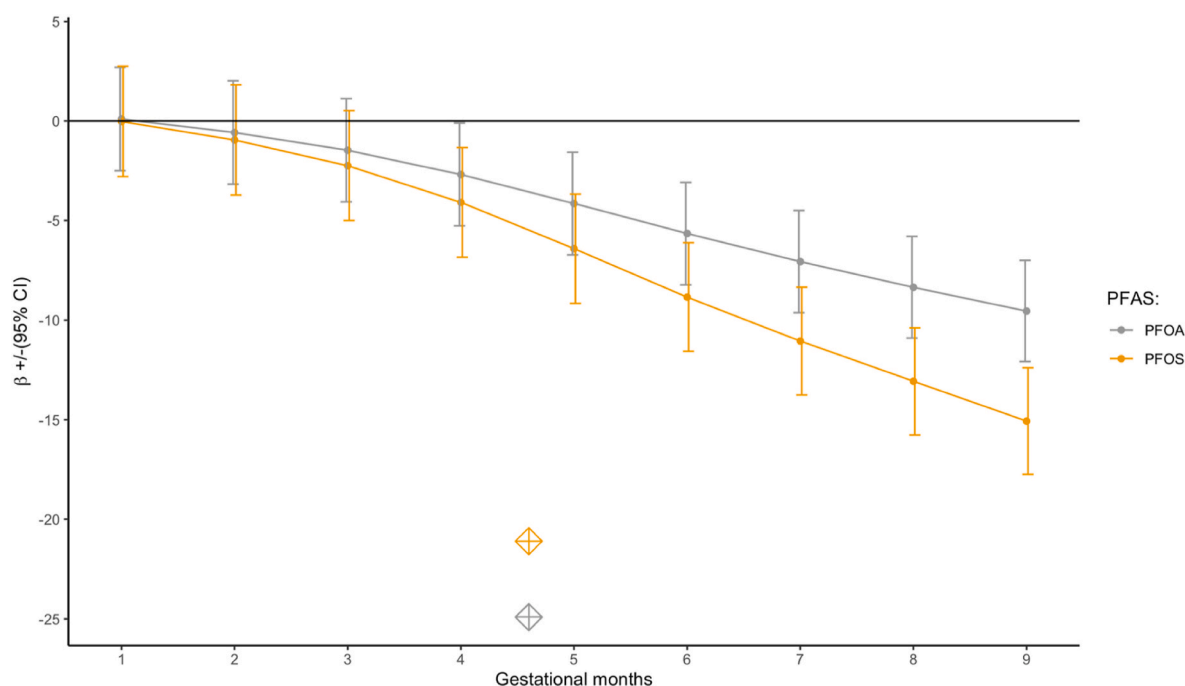


Fig. 3. Association between simulated maternal plasma PFOA and PFOS levels and birth weight due to glomerular filtration rate using PBPK model according to timing of PFAS measurement (Excel Table S10). Diamonds represent betas of association between birth weight and measured PFOA (β [95 % CI] = -24.90 [-95.76 to 45.96] g) or PFOS (β [95 % CI] = -21.10 [-85.74 to 43.55] g) concentrations in the study (Excel Table S8). Abbreviations: PBPK: Physiologically Based Pharmacokinetic Model, PFAS: poly- and perfluoroalkyl substances, PFOA: Perfluorooctanoate, PFOS: Perfluorooctane sulfonate. Note: Due to the wide range of the confidence intervals of association between birth weight and measured PFOA and PFOS, the confidence intervals were not represented.

during pregnancy, which might modify PFAS effects (Morken et al., 2014; Braun et al., 2022; Zhang et al., 2023b). However, we used PBPK modeling to evaluate the impact of GFR on the PFAS-birth weight associations, and estimated that only a small proportion of the association may be attributable to confounding, in line with previous studies (Taibl et al., 2023; Manzano-Salgado et al., 2017). Moreover, our results suggest that the portion of the PFAS-birth weight association attributable to confounding by GFR increases as gestational age advances, in line with a meta-analysis (Steenland et al., 2018). The cluster-based PFAS analysis comprehensively examines the mixture effects of PFAS. The longitudinal design provided numerous growth measurements per individual, varying in timing. Besides growth parameters at birth, we used repeated ultrasound measures to track fetal development throughout pregnancy. It is worth noting that different medical staff conducted these measurements during routine monitoring, which may have introduced variability and potential measurement errors. This, along with our limited sample size, could have bias our results toward the null and thus impacted the detection of associations related to fetal growth. For postnatal growth, the Jenss-Bayley prediction model allowed us to predict growth parameters at the same age for all children, accounting for repeated measurements and potential missing data (Botton et al., 2014). Finally, we investigated the impact of prenatal PFAS exposure on multiple growth measurements from the second trimester to 3 years, using both classic growth and adiposity measurements, providing a broad picture of early life exposure effects.

In conclusion, in this prospective cohort study, higher maternal serum PFAS concentrations were associated with increased head measurements and BMI velocity on childhood, which are potential determinants of obesity in later-life.

CRediT authorship contribution statement

Marion Ouidir: Writing – original draft, Formal analysis. **Marc-André Verner:** Writing – review & editing, Methodology, Formal analysis. **Valérie Siroux:** Writing – review & editing, Resources. **Sarah**

Lyon-Caen: Writing – review & editing, Project administration. **Aminata H. Cissé:** Writing – review & editing, Data curation. **Jérémy Botton:** Writing – review & editing, Resources. **Barbara Heude:** Writing – review & editing, Resources. **Séverine Valmary-Degano:** Writing – review & editing, Resources. **Line Småstuen Haug:** Writing – review & editing, Resources. **Cathrine Thomsen:** Writing – review & editing, Resources. **Sam Bayat:** Writing – review & editing, Investigation. **Rémy Slama:** Writing – review & editing, Funding acquisition. **Claire Philippiat:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2025.121814>.

Data availability

Data will be made available on request.

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